

ABDELLAH BOUGATAYA

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EDUCATION

Yıldız Teknik Üniversitesi

September 2023 - July 2027

Control and Automation Engineering

CGPA: 3.34 / 4.00

Electronics Academic Team Leader, Inonest Community; Student Affairs, International Students Club

PROJECTS

Remote Headless UAV Simulation & Telemetry Server

February 2026 - May 2026

Headless SITL simulation server with low-latency streaming and telemetry routing

- Deployed a headless ArduPilot SITL and Gazebo simulation environment on a remote server to offload heavy computational physics and visual rendering from the development machines.
- Engineered a low-latency video streaming pipeline using GStreamer and MediaMTX, and configured a MAVLink router over TCP/UDP to manage bidirectional telemetry synchronization.

Custom Flight Controller (AVIONX team)

April 2025 - July 2025

Bare-metal flight controller firmware with cascaded PID control and motor system identification

- Modeled quadcopter dynamics using Newtonian physics and system identification techniques, developing cascaded PID control loops in MATLAB & Simulink to achieve a **10% overshoot and a 0.5-second settling time**.
- Translated continuous-time models into bare-metal C/C++ firmware running on an ESP32 microcontroller, optimizing memory layout and execution speed by utilizing custom structs, pointers, and modular software files.

Double Copter Self-Balancing Aero-Pendulum

January 2025 - April 2025

Bi-actuated self-balancing pendulum system with sensor fusion and software filtering

- Implemented a software low-pass filter to reject sensor noise and deployed a complementary filter strategy for fast, low-overhead state estimation. Integrated a backward-difference derivative method and dynamic clamping anti-windup for the PIV controller.
- Developed a modular firmware library for the BMI160 IMU from scratch, implementing low-level I²C communication protocols over Wire.h to log real-time sensor data.

Rocket Engines API

June 2024 - July 2024

Backend REST API for rocket engine metrics with automated data scraping

- Wrote a custom Python script to scrape, parse, and clean non-uniform database tables from web data, resolving interval values using median statistical approximations.
- Developed a high-performance REST API using Golang to query and serve processed rocket engine metrics from an SQLite database via JSON endpoints on port 8080.

SKILLS & CERTIFICATIONS

Control & Estimation
Software & Embedded
Tools & Platforms
Languages
Certifications

System Identification, Kalman Filters, LQR/LQG, PID Control, Signal Filtering
C, C++, Python, Golang, SQL, ROS2, L^AT_EX, Embedded Firmware, I²C, MAVLink
MATLAB & Simulink, Git, Linux, ArduPilot SITL, Gazebo, SOLIDWORKS
Arabic (Native), English (Advanced), French (Advanced), Turkish (Intermediate)
CS50: Introduction to Computer Science (Harvard/edX), ROS2 for Beginners,
MATLAB Fundamentals, SOLIDWORKS CAD Fundamentals

REFERENCES

Research student under Assoc. Prof. Claudia YAŞAR at the Control Tech Lab (YTU). Academic and professional references are available upon request.